Hipergator and command line basics

Phylogenomics

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1. Prior to this lab you should have done the following:

1. Obtained a username and password for Hipergator
2. Installed the following software on your laptop:
   - Terminal (Mac) or PC equivalent
   - Cyberduck (Mac) or PC equivalent (e.g., Putty)
   - TextWrangler (Mac) or PC equivalent (e.g., Notepad)
3. Navigate to `/ufrc/zoo6927/share/hpg_basics`, and copy the files there to your own folder
4. Configured Cyberduck (or equivalent) to work with Hipergator

2. Getting onto the server

Open Terminal or PC equivalent (Putty, etc.). At the prompt (the prompt is the $; you do not enter this, it just appears), type the following, where `<user>` is your username. N.B.: In general, where I put these carrot symbols: `< >` it means that the text between them is something variable, sometimes something specific to you; in this case, `<user>` means that you enter your username. Enter your password when asked; it will NOT appear on the screen as you type it. Type carefully; it will lock you out after three incorrect attempts. You can hit delete a bunch of times to erase all the keystrokes you've made if you think you have made a mistake, and start over.

```
$ ssh <user>@hpg2.rc.ufl.edu
```

Once you are logged on, you will see something like this:

```
[user@gator2 ~]$
```

This tells you that you are now on Hipergator (HPG). Before you continue, you need to do another step to move to the appropriate place on HPG. By default, the place you land when you logon is your home folder, but this is not actually where you want to work. To change to the correct location, you will need to know the group that you're part of, and you'll use the 'cd' function to change directory (that's what cd stands for) to move there. First type 'pwd', which is the command to check your present working directory. You should see something like this:

```
$ pwd
/home/emilysessa
```

This is the location where we don't want to be. To get to the right place, enter the following:

```
$ cd /ufrc/<group>/<user>
$ pwd
```

The last command should now display exactly the location you entered with the `cd` command.

`pwd` is "present working directory" - this command will always give you your current location, i.e., the folder that you are working in at the moment. You can move around between directories easily, using `cd`, the command above where you moved from the home folder to the scratch folder (`cd` stands for "change directory").

3. Basic useful unix commands

Once you are on the server, there are a number of helpful commands that you should memorize, and will find yourself using frequently. These are demonstrated in the code below, which has comments indicated with #.

```
$ mkdir Basics  #this command will create a new directory (folder) called Basics
$ cd Basics  #this will move you into the Basics directory
$ cd ..  #the directory above where you currently are is called .. and this will move you back to it
$ ls  #this will display a list of the current directory's contents
$ ls -l  #this displays the directory content list with additional information
```

In addition to `..` being the directory immediately above you, `.` is the directory where you currently are. This is useful to know and will come in handy later.

4. Getting files onto hipergator

There are many ways to move files between your local machine and the server; these include commands like `push`, `wget`, `scp`, and many others. I am lazy and like to drag and drop. I also like to be able to view my files more conveniently than as a list on the server. This is a personal preference, and both of these can be accomplished using an FTP program (I use Cyberduck), which you should already have installed. One useful modification to make to Cyberduck's setup is to make it such that if you double click a file name, it will open that file using your preferred text editing software, directly on the server. The default is to have double clicking download the file, which gets really annoying and cumbersome. To change this, do the following: Select Preferences from the Cyberduck menu, and select Editor. Choose your editor of choice from the dropdown menu, and hit the button for "always...". Now choose the Browser menu option, and check the box next to "Double click opens file in external editor". You're all set!

In Cyberduck, navigate to the new Basics directory you created above, and drag "dummy.txt" from wherever it is on your laptop to that directory. It should transfer automatically. Now we can practice a few more commands...

5. Other useful commands

First, `pwd` again to check on where you are. You should be in `/ufrc/<group>/<user>/Basics`. If not, get there by using `cd`. Then use `ls` to list the files there and doublecheck that "dummy.txt" shows up in the list.

```
$ cp dummy.txt ..  #this will copy "dummy.txt" from the current directory to the directory above
$ cp dummy.txt dummy1.txt  #this will make a duplicate of dummy.txt called dummy1.txt
$ mv dummy1.txt ..  #this will move (not copy!) the file to the directory above
$ cd ..  #move yourself to the directory above Basics
$ rm dummy.txt  #this will delete the file IMMEDIATELY - THERE IS NO UNDOING THIS!!!!!!!
```

A note on the `rm` command: there is no UNDO function for this. If you delete a file, or a directory, it is gone, immediately, forever. There is no recovery. Be extremely certain about what you're doing before you delete anything - this applies x1000 to deleting directories, which are entire folders that you then cannot recover!

Often the files we work with will be too long to display on screen, but seeing the first or last few lines can be very helpful:

```
$ head dummy.txt  #displays first 10 lines
$ tail dummy.txt  #displays last 10 lines
$ wc -l dummy.txt  #tells how many total lines are in the file; instead of -l, you can use
-w to give words, -c to give characters, or a combination of these, e.g., -lc
```

As you probably noticed, that file had the word "dummy" in it quite a few times. We can tell exactly how many times using a standard unix command called `grep`, and various options, or flags, that you can use with it.

```
$ grep dummy dummy.txt  #prints out each line that contains the word dummy, in the file dummy.txt
$ grep -i dummy dummy.txt  #the -i flag makes the search case-insensitive (sensitive by default)

Other options:
-v display those lines that do NOT match
-n precede each matching line with the line number
-c print only the total count of matched lines

You can combine these, e.g.:
$ grep -vc dummy dummy.txt  #this prints out the number only, for the lines that do NOT match
```

6. Tab to complete

Once of the most useful tips for working on the command line is to use the Tab key to auto-complete file and directory names as you type. This is an essential trick, and you should get used to using tab extensively. Besides Enter, it should be the key you hit the most often. Try this:

```
$ cp du  #now hit the Tab key. "dummy.txt" should autofill on the command line.
$ cp di  #hit Tab again. This time nothing happens (or it will flash or make a noise).
```

This is a beautiful tool for several reasons: first, it saves you time from having to type out a long line, if you want to reach a file that is several directories away from you. Second, if the file you are searching for is not actually in the directory where you are located, or you are typing the name wrong, you will figure it out very quickly, because pressing Tab will produce no results. This prevents you from typing out a whole long file address, only to discover the file is not actually present in the folder where you are located, or that you made a type way back at the beginning. Use tab constantly!.

7. Editing text directly on HPG with nano

Nano is a text editor built directly into unix. There are other such programs, including vim, but nano is one of the most user-friendly. It is useful if you don't want to use Cyberduck to edit a text file using Textwrangler.

```
$ nano dummy.txt  #this will open the file in nano, in the same terminal window.
Did you use tab???
```

You can edit the file as you like. On a mac, ctrl-X closes it. Type 'Y' and then 'enter' to save it to the appropriate file name (you can save it as something different at this point if you wish), and then close it.

8. Filetypes in phylogenetics/phylogenomics, and some practice

Nexus - very common (used in PAUP, MrBayes, etc.)

Find the primates.nex file on Canvas and transfer it to Hipergator. Use `nano` to view the file. Note that the file is composed of "blocks"; each block starts with the word `begin <block type>;` and somewhere later (sometimes many lines later), ends with `end;`. Also note that every single individual line ends with `;`.

1. How many taxa are in here and how can you tell without counting the lines in the matrix? How many characters are in the matrix?
2. How many times does the sequence motif "ACCTCGCTCCAAT" appear in primates.nex?
3. Which species have the same sequence motif, "ACCTCGCTCCAAT", in the alignment?

The remaining are optional things to try if you want to, and don't require you to use HPG.

Newick - a commonly used tree format

Download the file primates.tre to your laptop. Open the file in FigTree, and play with the options in the left sidebar. You can use the File > Export Trees... function to export the file in Nexus (or Newick) format, though the original file is already in Newick format. Note that in order for Figree to open files, they usually need to end with `.tre`.

Now open the original file in TextWrangler. Use the graphical representation in Figtree and the text version in TextWrangler (both derived from the Newick-formatted file) to answer these questions.

4. How long is the branch that subtends the clade including Homo_sapiens, Pan, and Gorilla?
5. What is the appropriate section of Newick code that defines this clade? You can either find it and copy it out of the Newick file, or try to write it yourself by looking at the figure.

Phylip - used with RAxML, among other things

Download the primates.phy file to your laptop, and open it with TextWrangler. Notice the differences between this and the Nexus file.